

Field Analysis of Mobile and Self-Pay Applications at IOI City Mall

1. Introduction

This report presents an analysis of one mobile application (Grab) and three self-payment systems (Lotus self-checkout, Decathlon self-checkout, and AEON self-checkout) used at IOI City Mall. The aim is to evaluate these systems based on key usability criteria and propose relevant improvements. This project was conducted by a group of three students as part of the CSC1301 course requirements.

2. Apps and Systems Overview

Grab App: A ride-hailing app used to book transportation. It allows users to choose ride types, view estimated fares, and track drivers in real-time.

Lotus Self-Checkout: A self-service machine where customers scan, pay, and bag their own groceries without needing a cashier.

Decathlon Self-Checkout: A self-checkout system used in the sports store Decathlon to scan and pay for sports items like bottles or clothes.

AEON Self-Checkout: A supermarket self-checkout system allowing customers to complete their grocery shopping independently using digital payment options.

3. Usability Analysis

3.1 Grab App

Usability Criteria	Analysis
Learnability	Very easy to learn; steps are clear and guided.
Efficiency	Booking is fast and takes less than 2 minutes.
Memorability	Easy to reuse after a gap due to consistent UI.
Errors	Minimal errors; easy to fix if wrong destination is entered.
Satisfaction	Highly satisfying with smooth booking and real-time tracking.

Accessibility	Moderate; lacks features for visually impaired users.
Responsiveness	Very responsive with no app crashes.
Aesthetic and Minimalist Design	Modern, clean, and user-friendly design.

3.2 Lotus Self-Checkout

Usability Criteria	Analysis
Learnability	Moderate; might need assistance on first use.
Efficiency	Efficient after initial learning curve.
Memorability	Takes a little time to remember the steps.
Errors	Possible errors in scanning or payment selection.
Satisfaction	Satisfying once familiar.
Accessibility	Moderate accessibility support.
Responsiveness	Good responsiveness, some lag at payment.
Aesthetic and Minimalist Design	Simple design but slightly cluttered screen.

3.3 Decathlon Self-Checkout

Usability Criteria	Analysis
Learnability	Easy and intuitive for most users.
Efficiency	Very quick checkout process.
Memorability	Easy to remember steps.
Errors	Rare and recoverable.
Satisfaction	Highly satisfying experience.
Accessibility	Limited accessibility features.
Responsiveness	Smooth and responsive.
Aesthetic and Minimalist Design	Clean and simple design.

3.4 AEON Self-Checkout

Usability Criteria	Analysis
Learnability	Very easy to use even for first-timers.
Efficiency	Fast and simple checkout.

Memorability	Steps are memorable and consistent.
Errors	Few scanning issues but manageable.
Satisfaction	Comfortable and convenient.
Accessibility	Offers basic accessibility support.
Responsiveness	Good responsiveness overall.
Aesthetic and Minimalist Design	Simple and organized layout.

4. Comparison and Contrast

Below is a comparison table of the three self-payment systems based on usability criteria.

Criteria	Lotus	Decathlon	AEON
Learnability	Moderate	Easy	Very Easy
Efficiency	Moderate	High	High
Memorability	Needs re-familiarization	Easy to resume	Easy to resume
Errors	Some confusion	Few and recoverable	Minor scan errors
Satisfaction	Average	High	High
Accessibility	Moderate	Limited	Moderate
Responsiveness	Good	Very Good	Good
Aesthetic and Minimalist Design	Slightly cluttered	Clean	Simple

5. Suggested Improvements

- Grab App:
 - Add accessibility support for visually impaired users.
 - Introduce a live help/chatbot for troubleshooting.
 - Improve map directions to parking zones or pickup spots.
- Lotus:
 - Add cancel or undo options during scanning.
 - Provide clearer step-by-step instructions on screen.

- Station a staff member near machines for first-time help.
- Decathlon:
 - Improve signage and instructions near the checkout.
 - Add a voice guide for visually impaired users.
 - Simplify the payment confirmation button.
- AEON:
 - Add multi-language support for tourists.
 - Improve barcode scan accuracy.
 - Expand bagging area for convenience.

6. Conclusion

The field visit and analysis showed that mobile and self-payment systems offer great convenience, efficiency, and independence. However, improvements in accessibility, guidance, and interface design can make them even better. Through this hands-on experience, we gained a deeper understanding of real-world usability and user experience challenges.

Video Link

https://youtu.be/evYt_ygNBZY